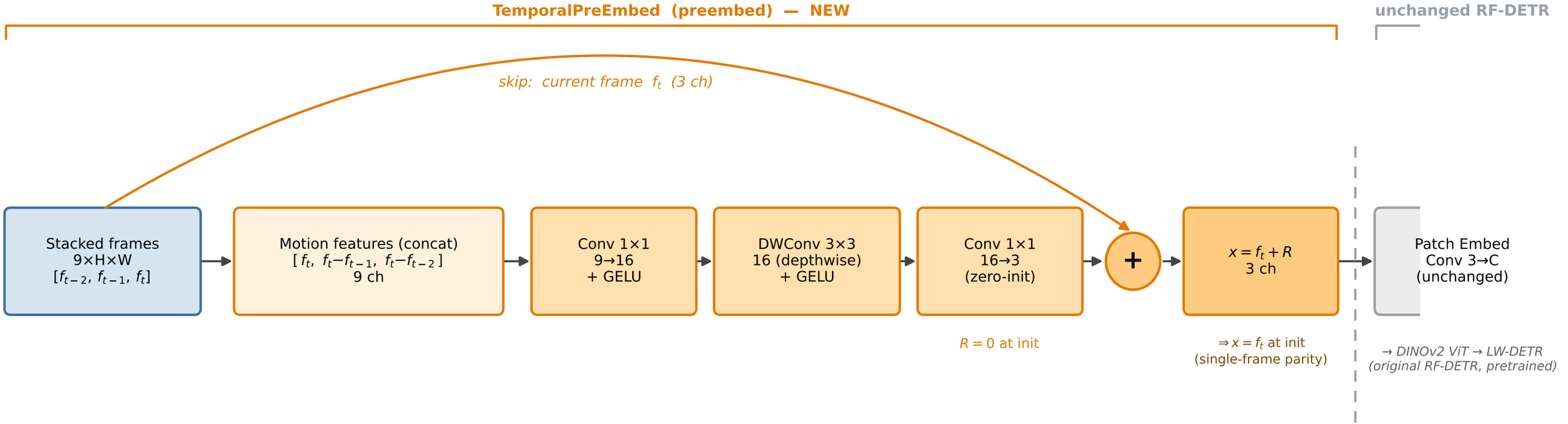
TemporalPreEmbed (preembed) — NEW
unchanged RF-DETR
skip: current frame f_t (3 ch)
Stacked frames
9×H×W
$[f_{t-2}, f_{t-1}, f_t]$
Motion features (concat)
$[f_t,\ f_t - f_{t-1},\ f_t - f_{t-2}]$
9 ch
Conv 1×1
9→16
+ GELU
DWConv 3×3
16 (depthwise)
+ GELU
Conv 1×1
16→3
(zero-init)
+
$x = f_t + R$
3 ch
Patch Embed
Conv 3→C
(unchanged)
$R = 0$ at init
$\Rightarrow x = f_t$ at init
(single-frame parity)
→ DINOv2 ViT → LW-DETR
(original RF-DETR, pretrained)